



PYTHON Programming

Conditional and Iterative Statement

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- When program uses a condition to decide which set of statement is to be executed between alternative set of statement, is called Conditional statement.
- Python provide following type of conditional construct:
 - If Statement
 - If else statement
 - If elif else statement
 - Nested if else statement

➤ The statement if tests a particular condition. Whenever that condition evaluates to true, an action or a set of action is executed. Otherwise, the action are ignored.

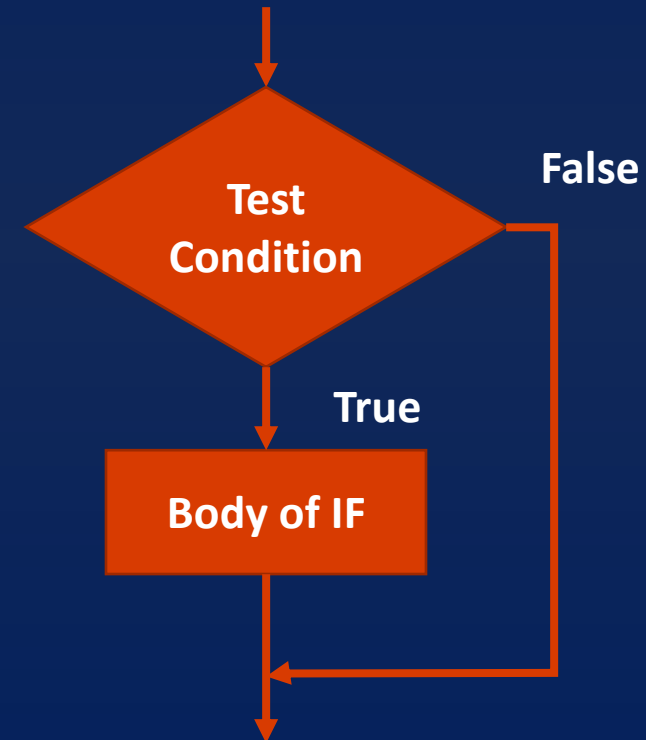
➤ Syntax:

if **condition**:

Statement 1

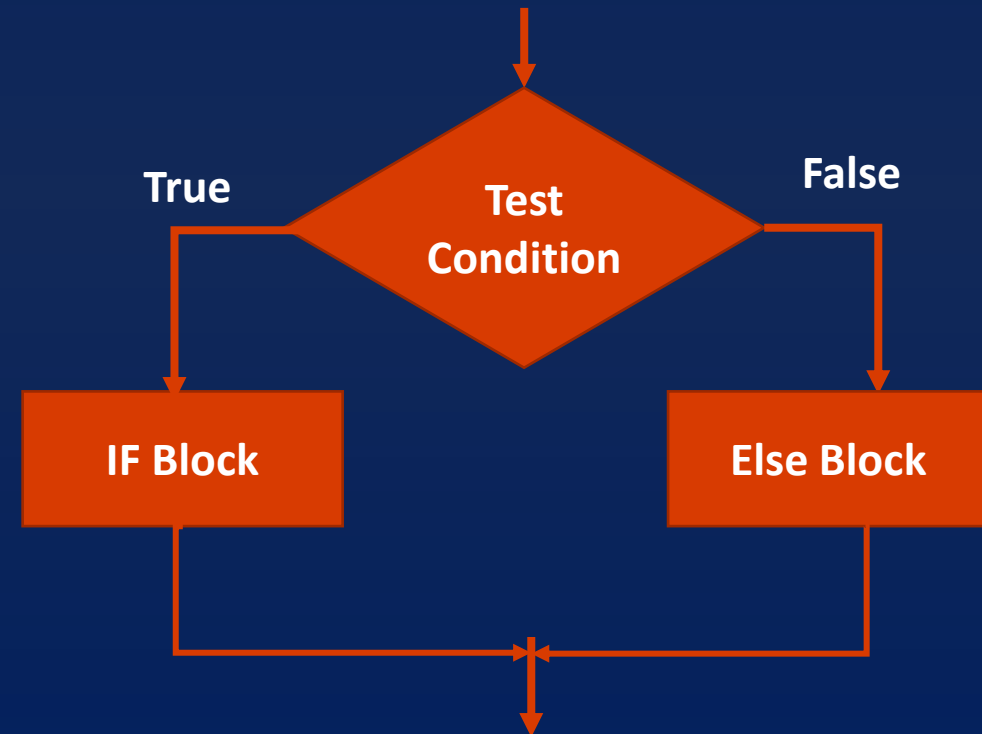
Statement 2

.....



- The if..else statement evaluates test expression and will execute the body of if only when the test condition is True.
- If the condition is False, the body of else is executed. Indentation is used to separate the blocks.
- Syntax:

```
if condition:  
    Statement  
    .....  
else:  
    Statement  
    .....
```



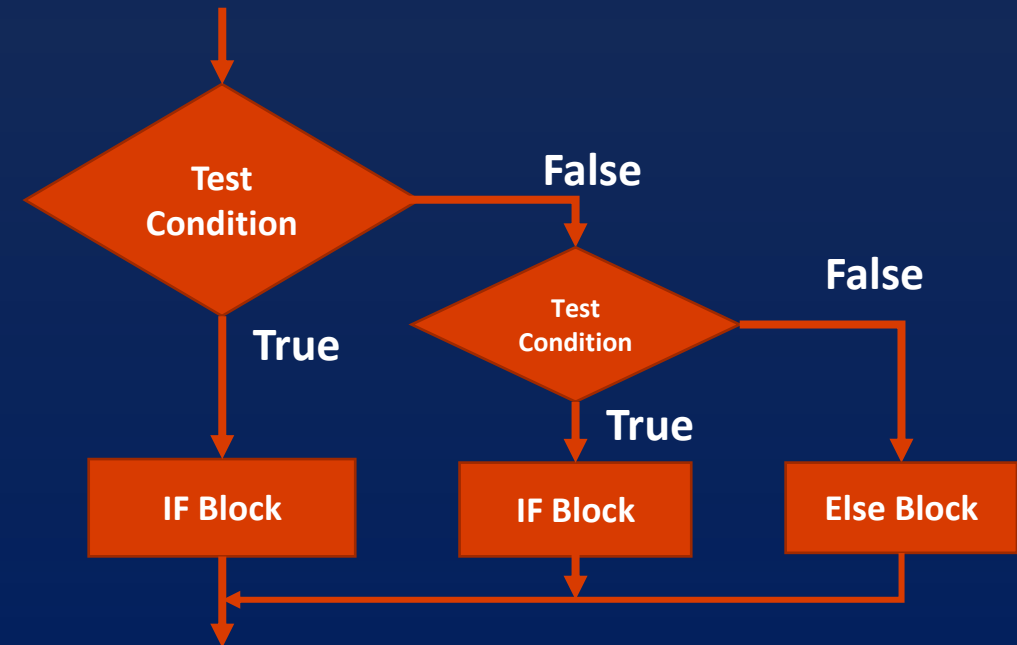
- A dartboard of radius 10 units and the wall it is hanging on are represented using a two-dimensional coordinate system, with the board's center at coordinate (0,0). Variables x and y store the x-coordinate and the y-coordinate of a dart that hits the dartboard. Write a Python expression using variables x and y that evaluates to True if the dart hits (is within) the dartboard, and then evaluate the expression for these dart coordinates:
 - (a) (0,0)
 - (b) (10,10)
 - (c) (6,6)
 - (d) (7,8)

IF ELIF ELSE Statement

- The elif is short for else if. It allows us to check for multiple expressions.
- If the condition for if is False, it checks the condition of the next elif block and so on.
- If all the conditions are False, the body of else is executed.
- Only one block among the several if...elif...else blocks is executed according to the condition.

➤ Syntax:

```
if Condition:  
    Statement  
elif Condition:  
    Statement  
else:  
    Statement
```



Nested IF ELSE Statement

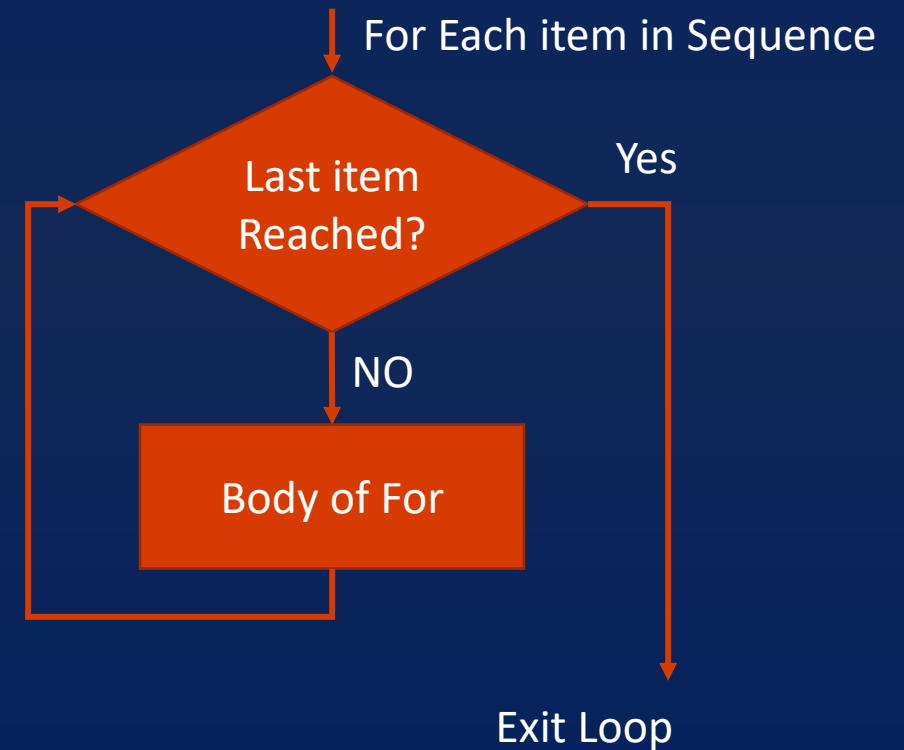
- We can have a if,if else or if...elif...else statement inside another if, if else or if...elif...else statement. This is called nesting in computer programming.
- Any number of these statements can be nested inside one another. Indentation is the only way to figure out the level of nesting.
- Syntax:
 - if **condition**:
 - if **condition**:
 - Statement or another if Statement
 - else:
 - Statement or another if Statement
 - else:
 - Statement or another if Statement

- Loops are used in programming to repeat a specific block of code.
- A loop is defined as a block of instructions repeated for a certain number of times.
- Python provide two type of Loop construct:
 - for Loop
 - while loop

➤ The for loop in Python is used to iterate over a sequence (list, tuple, string) or other iterable objects. Iterating over a sequence is called traversal.

➤ Syntax:

***for var in sequence:
loop body***



range() Function



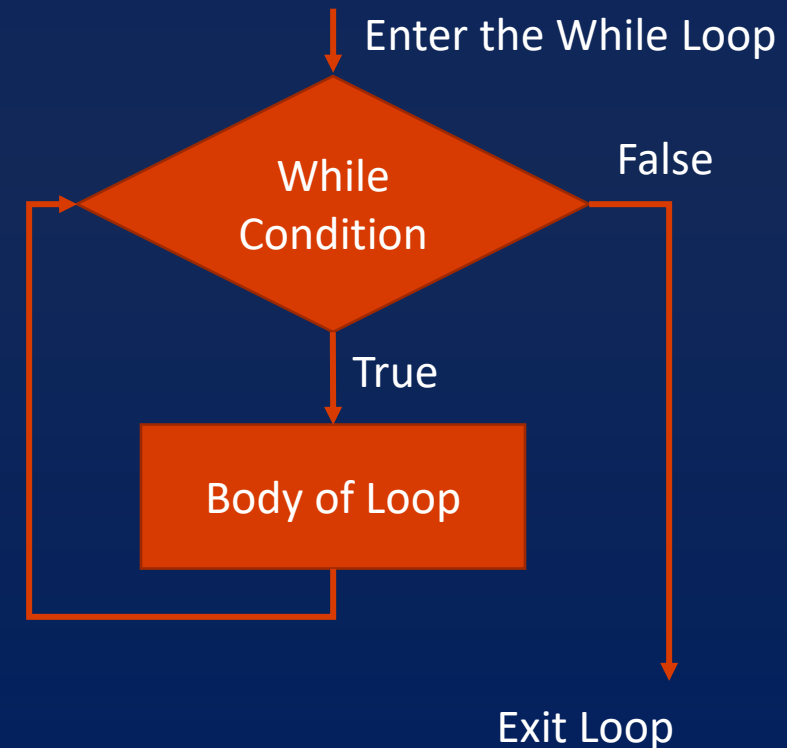
- We can generate a sequence of numbers using range() function. range(10) will generate numbers from 0 to 9 (10 numbers).
- We can also define the start, stop and step size as
 - range(start, stop, step_size)
- step_size defaults to 1 if not provided.

- A for loop can have an optional **else** block as well. The else part is executed if the items in the sequence used in for loop exhausts.
- The break keyword can be used to stop a for loop. In such cases, the else part is ignored.

```
n=int(input("Enter any Number : "))
for a in range(2,n):
    if n%a==0:
        print("No is Not Prime")
        break
else:
    print("No is Prime")
```

- The while loop in Python is used to iterate over a block of code as long as the test expression (condition) is true.
- We generally use this loop when we don't know the number of times to iterate beforehand.
- Syntax:

while test_expression:
Body of while



- A loop may also contain another loop within its body.
- This form of a loop inside another loop is called **nested loop**.
- In a nested loop construction, the inner loop must terminate before the ending of outer loop.
- Example:

```
for a in range(1,6):  
    for b in range(1,4):  
        print("*",end="")  
    print()
```

- Those statement which transfer the program control, unconditionally within the program, from one place to another called **jump** statement.
- **break** Statement
- **continue** Statement
- **pass** Statement

- The break statement terminates the loop containing it. Control of the program flows to the statement immediately after the body of the loop.
- If the break statement is inside a nested loop (loop inside another loop), the break statement will terminate the innermost loop.
- Syntax:

break

➤ The continue statement is used to skip the rest of the code inside a loop for the current iteration only. Loop does not terminate but continues on with the next iteration.

➤ Syntax:

continue

```
for val in "string":  
    if val == "i":  
        continue  
    print(val)  
print("The end")
```


- In Python programming, the pass statement is a null statement. The difference between a comment and a pass statement in Python is that while the interpreter ignores a comment entirely, pass is not ignored.
- Syntax:
pass

THANK YOU

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